26F Stat TA Session W7

💡 Midterm Information:

- time
- No questions about `R` will appear in midterm exam!
- No TA session in W8
- Email me to arrange the office hour if you need it!

0. Preparation for TA Session

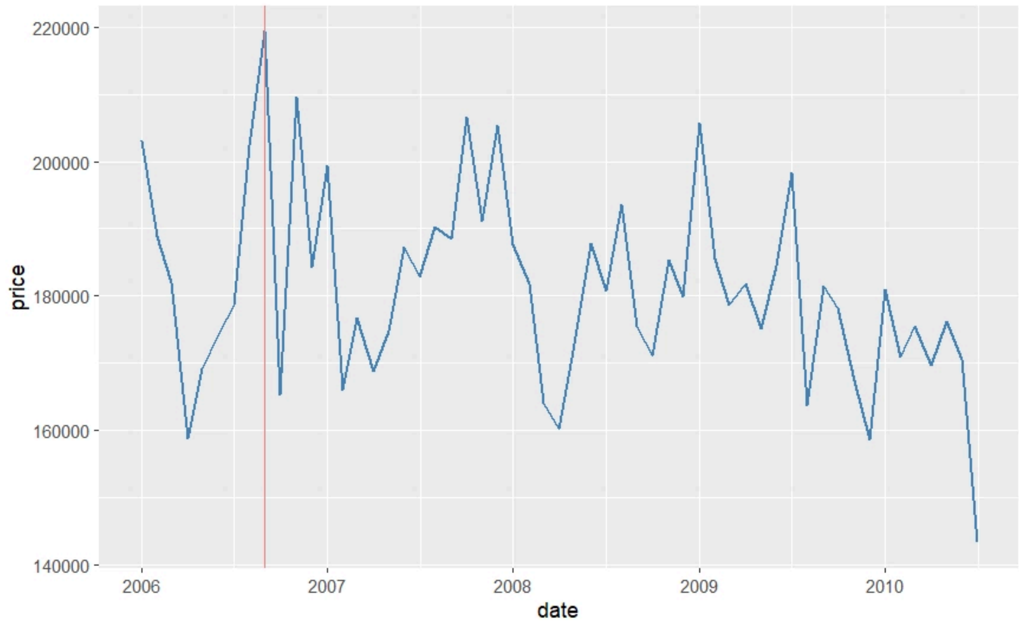
- Download the sample code and attached datasets: `ames` & `loans_full_shema` click here
    - The former appears in W6 and the latter in W3!
- Also import the datasets into your environment

1. Finishing the Course of Data Visualization

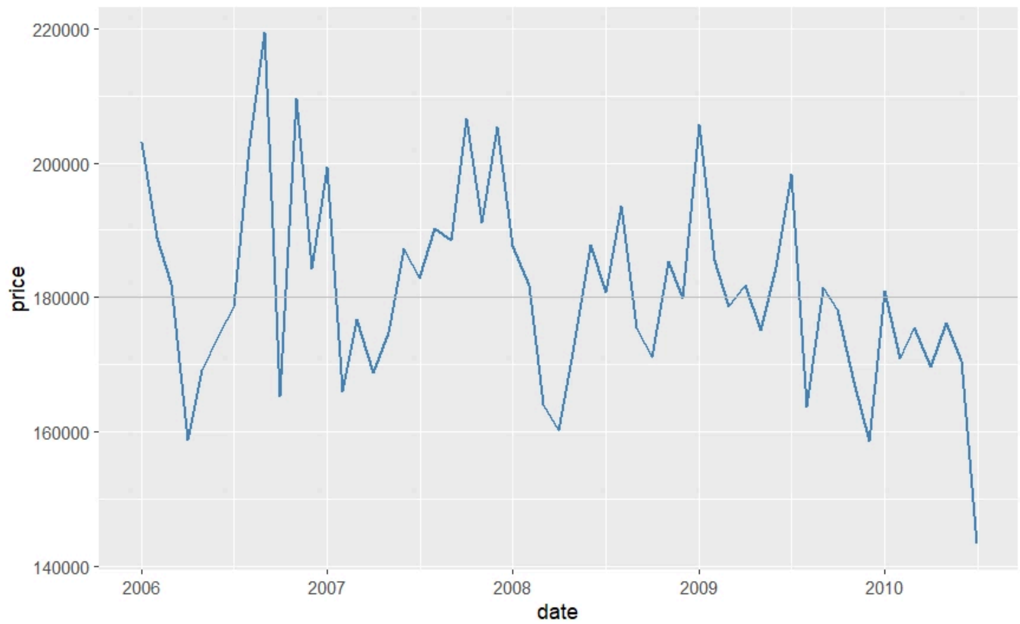
- Today we are going to introduce you to the last few things you should know when doing data visualization works!
- A well-selected and appropriate figure may be enough to see through the data, but sometimes adding some marks help others see what you see

Vertical Line/ Horizontal Line `geom_vline()` `geom_hline()`

```
ggplot(housing, aes(x = date, y = price)) + geom_line(stat = "summary", fun = "mean",
linewidth = 0.7, color = "steelblue" ) + # add vertical line to the date with the
highest average price geom_vline(xintercept = as.Date("2006-09-01"), color =
"lightcoral")
```




```
ggplot(housing, aes(x = date, y = price)) + geom_line(stat = "summary", fun = "mean",
linewidth = 0.7, color = "steelblue" ) + geom_hline(yintercept = 180000, color = "gray")
```



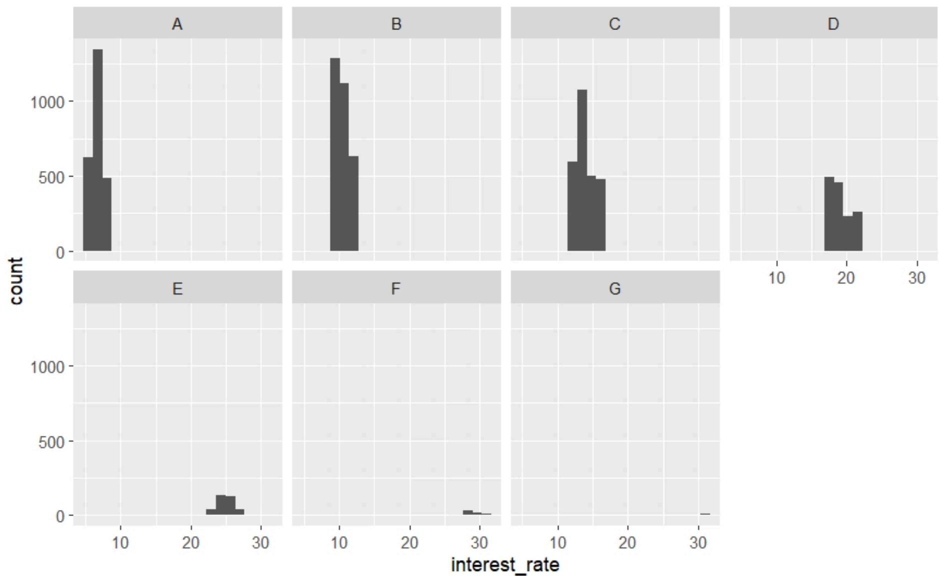
Split Figures into Groups `facet_wrap()`

- When dealing with grouped data, it is helpful to show them at the same time in smaller panels so that we can compare them directly!
- Scaling is the KEY! Use the fixed scaling for all facets unless you have a good reason not to!
  - For example, you want to show the trend for different groups instead of the exact numbers

```
ggplot(loan, aes(x = interest_rate)) + geom_histogram(bins = 20) + facet_wrap(~grade,
scale = "fixed", nrow = 2) # default: fixed scaling
```

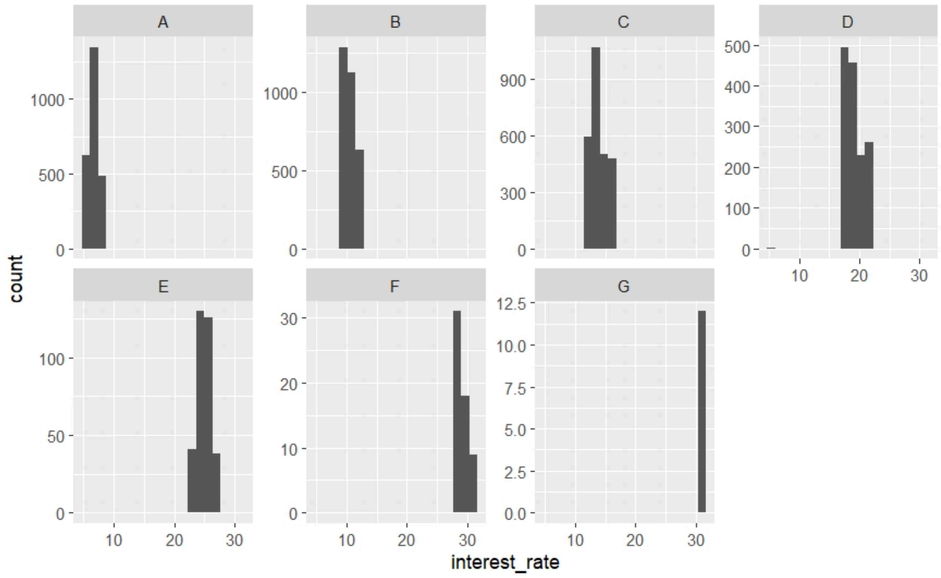
Syntax Decomposition

- `~grade` : facet by grade. Each unique grade level gets its own panel
    - This also works for multiple categories (e.g., `~grade + disbursement_method`)
- `scales =` : the default scaling is to use the fixed scaling for all panels. we can also set a different scaling method (e.g., `"free_y"` : fixed x-scaling and adjust y-scaling for each panel)
- `nrow` : place the panels in n rows
    - `ncol` also works



▸ Q. Should we use fixed scaling?

```
ggplot(loan, aes(x = interest_rate)) + geom_histogram(bins = 20) + facet_wrap(~grade,
scales = "free_y", nrow = 2) # y-axis automatically adjust scaling
```



`ggsave()`

- The function of `ggsave` is simple: save your plots!
- By default, it saves plots to the working directory. But you can also save in the plot folder!

Path

- There are 2 ways to set the saved path
  - Type the path inside `filename`
  - Use `path = ""`

```
ggsave(filename = "C:/Users/you/figures/vline_plot.png") ggsave(filename =
"figures/vline_plot.png") ggsave(filename = "vline_plot.png", path = "figures")
```

Customize your Plots

- One reason to use `ggsave()` (instead of simply click the produced plot and download it) is reproductivity
- Save plots in a specified format allows others to produce the exact same figure as we do
- By default, `ggsave()` saves the plot using the current plot size in `RStudio` . But we can customize the size and quality

```
ggsave(filename = "vline_plot.png", width = 6, # adjust the width height = 4, dpi = 300)
# adjust the quality (default = 300) # put all things together ggsave(filename =
"vline_plot.png", plot = vline_plot, path = "figures", width = 6, height = 4, dpi = 300)
```

Exercises

W7 Exercises